
From Listeners to Creators

**Bridging Musical Inequality through
Generative AI Music Tools**

**A Practical Study Based on the ADDIE Model in
Junior High School Education**

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1 Every Student Has a Song Inside — But a Wall Stands Between

The Old Reality

- Music composition has long required years of formal theory training
- Instruments take years to master
- Most students remain passive listeners
- Creation is a privilege of the formally trained few



Generative AI

- SUNO AI: text input → full song generation
- No sheet music, no instruments needed
- Creative threshold shifts from **technical mastery to conceptual expression**
- **But new barriers emerged**

2 "Conceptual Materialization Dilemma"

*Students have rich emotional ideas —
yet cannot translate them into precise AI prompts*

All-English Interface



SUNO operates entirely in English, creating a language barrier

Prompt Precision Required



Output quality depends on prompt specificity, creating a high cognitive threshold

Limited Typing Proficiency



Students with typing difficulties are effectively excluded

3 Research Context

18

Classes

496

Student Creators

505

Original
Compositions

May–Jun

2025

Semester 2

Research Setting

Public junior high school, in Southern Taiwan
Grades 7–8 | ages 12–14
Cross-disciplinary: Technology × Arts



Prior Experience

The same cohort had prior AI scaffolding experience from the preceding semester
→ familiar with human-AI workflows, reducing onboarding friction

Linguistic Outcomes



Compositions span **Mandarin, English, Korean, Japanese, Vietnamese, Arabic, and more** —
the scaffolding system effectively bridges virtually any linguistic divide



4 Four Theoretical Pillars

ADDIE Instructional Design Model (Branch, 2009)

Five systematic phases:
Analysis → Design → Development
→ Implementation → Evaluation



Scaffolding Theory & Cognitive Load (Wood et al., 1976; Sweller et al., 1998)

Temporary ZPD support + reduced
extraneous cognitive load =
freed creative resources



Flow Theory (Csikszentmihalyi, 1990; Wang & Liu, 2025)

- Autotelic, deeply absorbing, effortless concentration
- Flow directly predicts music learning outcomes



Self-Determination Theory (SDT) (Deci & Ryan, 2000; Xia et al., 2022)

Autonomy + Competence + Relatedness
→ intrinsic motivation emerges
and endures



5 Three-Tiered Cognitive Scaffolding System

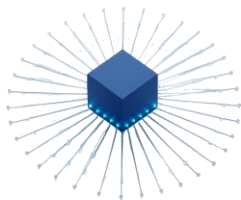
From "I have an idea" → to "I have a finished song"

Tier 1 · Thematic Framework

Creative Ideation Support

9 major themes
× 20 sub-topics
= 180 emotional starting points

→ Eliminates ideation paralysis



Tier 2 · Prompt Scaffolding Website

Operational Support

Google Sites
visual drop-down menus
language, genre, rhyme
one-click prompt copy

→ No English typing required



Tier 3 · Custom GPT

Linguistic Support

Mandarin input
→ any target language output

Automatic Mandarin
translation
romanized phonetics for other
languages



6 Tier 1: Thematic Framework

1 9 Major Themes — calibrated to adolescent emotional life

2 180 Creative Starting Points — each sub-topic is an immediate emotional anchor

3 Full Creative Agency — the framework gives direction, not a script

4 Why This Tier Matters — lowers the floor, never the ceiling

【 Fig. 1: 9×20 Thematic Matrix 】

 3. 我們的閃耀契約 此主題圍繞著朋友間的真摯情誼、共同經歷的冒險、互相扶持的溫暖以及對未來一起創造更多美好回憶的約定。	 8. 派對狂熱夜 此主題旨在描寫參與各種派對、慶典活動的熱鬧氛圍、音樂節奏、歡樂互動與難忘時刻。
3-1: 一起看過的星空	8-1: 舞池中央的雷射光束
3-2: 秘密基地的約定	8-2: 重低音震動的心跳
3-3: 吵過又和好的傻氣	8-3: DJ播放的下一首歌
3-4: 你懂我的奇奇怪怪	8-4: 跟著節奏一起搖擺
3-5: 一起挑戰的熱血沸騰	8-5: 夏日海灘的營火晚會
3-6: 交換日記裡的悄悄話	8-6: 沙灘排球與冰涼飲料
3-7: 因為你，我不再孤單	8-7: 夜晚城市的天台派對
3-8: 畢業後也要常聯絡喔！	8-8: 閃爍霓虹與都市夜景
3-9: 一起做的蠢事最難忘	8-9: 生日派對的驚喜尖叫
3-10: 我們的專屬暱稱	8-10: 彩帶、氣球與慶祝蛋糕
3-11: 生日時你給的驚喜	8-11: 復古迪斯科派對
3-12: 雨天共撐一把傘	8-12: 戴上面具的神秘舞會
3-13: 為你加油的吶喊聲	8-13: 露天音樂節的草地與星空
3-14: 一起追的夢想閃閃發光	8-14: 和朋友一起大聲合唱
3-15: 就算世界與你為敵，我挺你	8-15: 跨年派對的倒數時刻
3-16: 默契考驗：下一句你來接	8-16: 散場後微涼的街頭
3-17: 借我抄作業的恩情	8-17: 派對結束後耳邊的嗡嗡
3-18: 我們之間不用說謝謝	8-18: 回憶派對上的笑容
3-19: 老地方見的老習慣	8-19: 為了派對精心打扮
3-20: 這份友誼，無可取代	8-20: 狂歡之後的寧靜早晨
上一主題	主題 3 / 9
下一主題	主題 8 / 9
下一主題	下一主題

7 Tier 2: Prompt Scaffolding Website

- 1 Language selector
(Chinese / English / Japanese / Korean)
- 2 Theme-number input —
linked to the 9×20 matrix
- 3 Phonetic rhyme-scheme selector
- 4 Vocal arrangement dropdown
(solo / chorus / duet)
- 5 18 genre tags
(K-pop, Ballad, C-pop, EDM...)
- 6 One-click copy —
auto-generates a complete structured prompt

Development: ~2 weeks with Gemini collaboration; mobile-accessible.

【 Fig. 2: Prompt Scaffolding Website 】

SUNO歌詞提示詞產生器

主題總預覽

1. 心動的秘密信號

此主題旨在描寫青春期初次對人產生好感、暗戀時那種懵懂、害羞、酸甜又充滿秘密的心情與觀察。

1-1: 看見你就會心跳加速

1-2: 偷偷看你的側臉

1-3: 不敢說出口的「喜歡」

1-4: 為什麼在意你的每個小動作？

1-5: 收到你訊息時的開心

1-6: 擦肩而過的瞬間

1-7: 希望你能注意到我

1-8: 腦海裡都是你的笑容

1-9: 分享同一副耳機的午後

1-10: 猜不透你的心思

1-11: 練習本上偷偷寫下你的名字

1-12: 為了遇見你繞了遠路

1-13: 朋友起開時的害羞否認

1-14: 想像和你說話的場景

1-15: 你喜歡的歌我也偷偷聽

產生你的提示詞

語言: 中文 ▾

格式: 大主題編號~小主題編號 (例如 1-1, 5-18) ✎

主題 (輸入或點選左側編號): ✎

請參考上方主題描述，在此撰寫您想要的歌詞具體內容...

內容描述: ✎

這是您希望歌詞表達的具體情境或故事。

是否有參考歌詞: 否 ▾

押韻:

主要韻腳: 由你設計 ▾

次要韻腳 (可選): -- ▾

選擇您希望歌詞押的中文韻腳 (基於注音符號)。

歌手: 男 ▾ 演唱形式: 獨唱 ▾

例如: Mandopop, Ballad, Taiwanese pop... ✎

曲風 (請至少描述兩種風格): ✎

點選下方參考關鍵字可自動填入 (輸入框僅顯示英文):

Anison (動畫歌)

8 Tier 3: SUNO Lyrics Custom GPT

【 Fig. 3: Custom GPT 】



1 Student inputs creative intent in Mandarin

2 GPT outputs fully structured lyrics in the target language

3 Output includes structural markers: [Verse][Chorus][Bridge]

4 **Mandarin translation + romanized phonetics** automatically appended

5 **Self-authored lyrics work too:** adds only structural markers, never altering content

Supported languages:

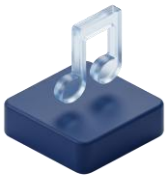
Mandarin, English, Japanese, Korean, Vietnamese, Arabic, and more

9 Six-Week Implementation (May–June 2025)

Weeks 1–3

Exploration & Foundation

- 8-part song structure (Intro→Outro)
- Analysis of 11 genres with demo tracks
- Rhyming techniques (Chinese & English)
- **3 self-designed instructional slide decks**
- SUNO account registration



Weeks 3–4

Scaffolded Creation

- **Select a theme** from the 9×20 matrix
- Generate prompts via the Scaffolding Website
- Paste into Custom GPT to receive lyrics
- Input into SUNO V4 Custom Mode to generate songs



Weeks 5–6

Refinement & Exhibition

- **Diagnose and correct AI** pronunciation errors
- Fix structural markers (/ → +)
- Advanced features: Cover, Extend, tempo adjust
- Padlet gallery + cross-class SUNO playlists



10 Results at a Glance

505

Original
Compositions

496

Student Creators

18

Class Playlists

Multilingual

Bridging any
linguistic divide

Compositions span Mandarin, English, Japanese, Korean, Vietnamese, Arabic, and more

3 Students Composed Entirely Self-Authored Lyrics

- Three students composed entirely self-authored lyrics; the Custom GPT added only structural markers
- → **evidence of genuine creative ownership** cultivated by the scaffolding

The Critical Role of Prior-Semester Experience

- The same cohort had completed an AI image-generation scaffolding module
- the preceding semester — **human-AI fluency enabled**
- the transformation from listener to creator



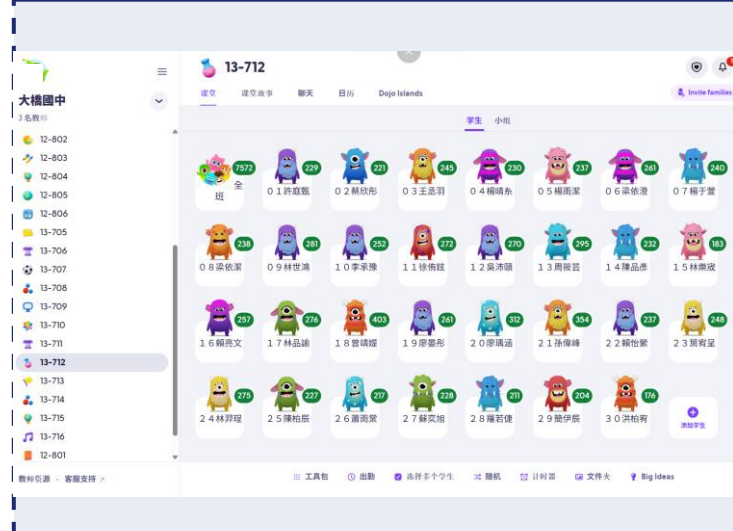
Assessment Strategy: Graded students' conceptual intention statements (not aesthetic quality) — construct-aligned with the research question.

11 Classroom Observation: Flow & Ecology

"The classroom transformed. Headphones on, creation began."

- All 18 classes — consistent behavioral patterns regardless of academic profile
- Behavioral disruptions virtually absent** — immersion replaced discipline (Weeks 3–6)
- Deep concentration — not from enforcement, but from authentic creative investment
- Flow no longer a privilege of the trained** — novices enter it too
- ClassDojo classroom management:
 - junior high students respond strongly to point-based systems
 - Flow likely benefits from intrinsic creative drive + established extrinsic incentives

【 Fig. 4: ClassDojo Classroom Management 】



12 505 Compositions — Bridging Any Linguistic Divide

Mandarin ~37%

English ~28%

Japanese ~18%

Korean ~14%

Vietnamese,
Arabic, and
others ~3%+

Student Works in the Padlet Gallery

"偷偷望一眼" — Mandopop, a young crush

"抽屜裡的夢" — Ballad, preserved memories

"一個人的自由" — a self-reliance theme

(**All students: zero prior music training**)

★ Graduation Song Projects

"最後一聲下課鐘" — The Last School Bell

"下一站,我們不在" — See You at the Next Stop

Entirely student-initiated, never prescribed.

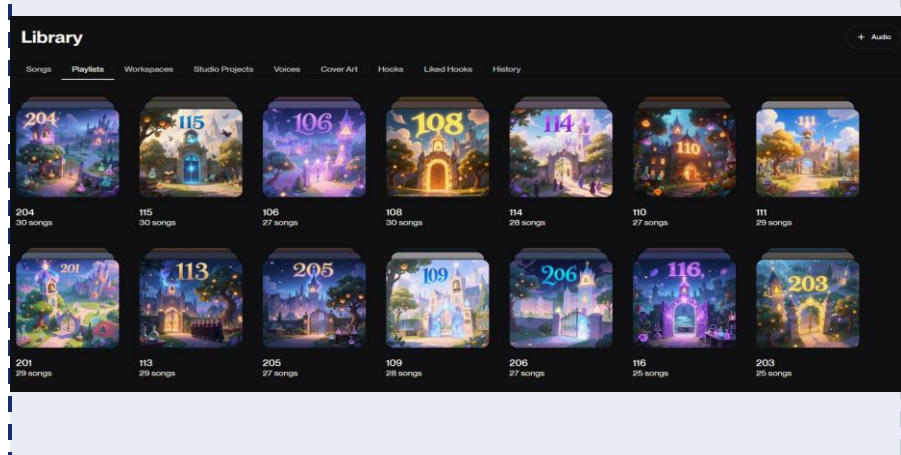
Faculty unanimously recognized these AI-assisted works as **artistically meritorious**.

13 Digital Gallery & SUNO Class Playlists

【Fig. 5: Padlet Digital Gallery】



【Fig. 6: SUNO Class Playlists】



Notable Finding: Music Magnet Class (20 students, formal music training)

"限界を超えて (Beyond the Limit)" · "정의하지 마 (Don't Define Me)" · "変わりたいのは私" · "The Way I See It"

Most engaged class of all 18 — works went beyond their classical training's framework.

Padlet: 496 student contributors · SUNO Library: 18 class playlists (25–30 songs each) · Total: 505 compositions

14 The Most Compelling Evidence

After the formal curriculum concluded — students kept creating.

Continued creating during lunch, after school, at home
Voluntarily investing personal time in music production

Graduation Song Projects
Entirely self-initiated;
faculty recognized the works as artistically meritorious

Three Layers of Creative Ownership
3 self-authored · Majority articulated rich intentions · Special-needs typed precise keywords
→ *Universal Design for Learning: agency takes different forms*

Composing in Languages Never Formally Studied
Mandarin translation + romanized phonetics
enable students to sing along

【 Fig. 7: Students in Flow 】



Self-Determination Theory (Deci & Ryan, 2000): all three psychological needs met → intrinsic motivation internalized, self-sustaining

15 Three Implications

Democratizing Aesthetic Creation

Flow can be democratized — novices enter it too.
The challenge-skill balance can be systematically engineered.



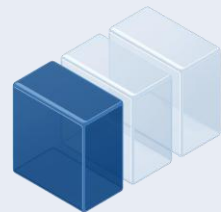
Educational Equity

The divide resides **not in hardware but in cognitive and linguistic resources**.
UDL principles aligned with SDG 4 (Quality Education) · SDG 10 (Reduced Inequalities)



Replicability — With Design-Cost Realism

Google Sites + Custom GPT: replicable, requiring ~2 weeks of design investment.
ADDIE is a **tool-agnostic, transferable blueprint**.
Prior human-AI literacy meaningfully accelerates adoption.



16 Future Outlook: From Classroom to Global Audiences

【 Fig. 8: Spotify 】



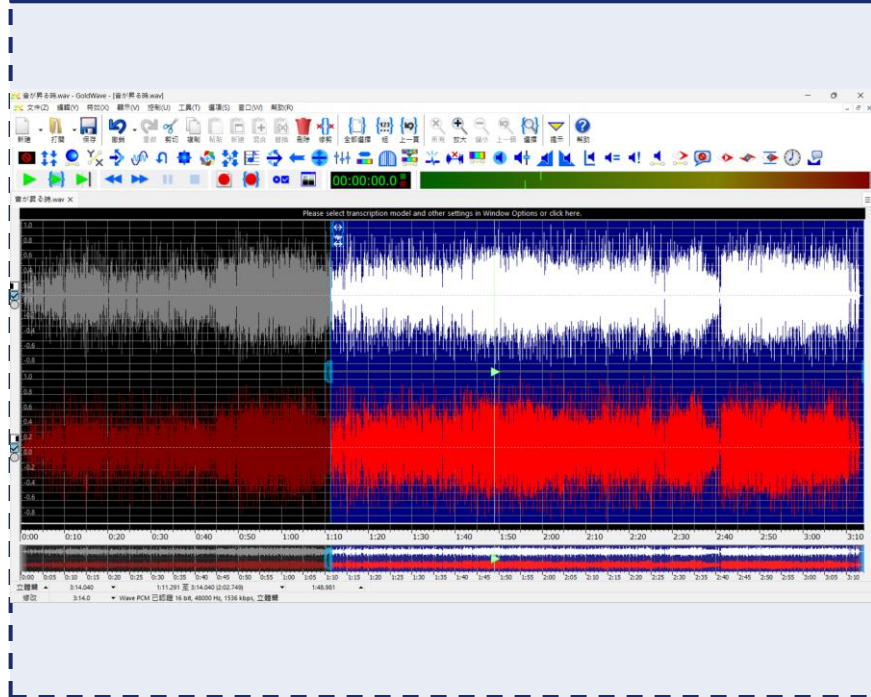
*Proof-of-concept:
the researcher distributed an AI-assisted composition to Spotify via Indiefy*

A Practical Workflow for Schools

- ① Institutional paid SUNO account
(free accounts: non-commercial exhibition only)
 - ② Teacher re-generates student works via
Remaster / Cover under the paid account
→ **commercially distributable audio**
 - ③ Post-produce with GoldWave to meet
Indiefy standards
 - ④ Allow ≥ 20 days for Indiefy review
-  Implementation evidence on the next slide.

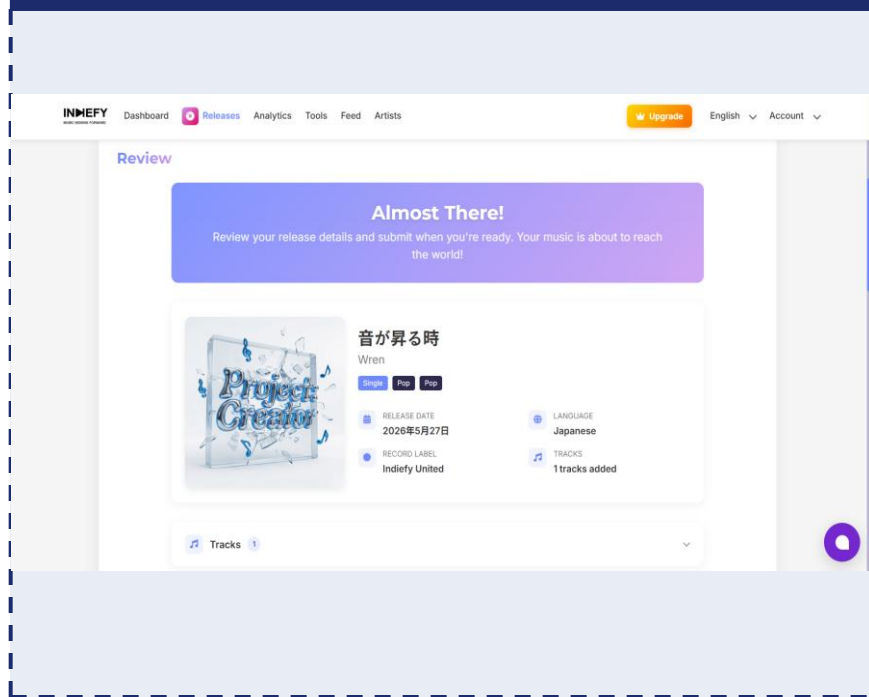
17 Implementation Evidence: Mastering & Distribution

【 Fig. 9: GoldWave 】



Audio Mastering

【 Fig. 10: Indiefy 】



Distribution

18 Conclusion

When the barriers recede — the music rises.

- The three-tiered scaffolding effectively bridges **cognitive, operational, and linguistic barriers**
- The ADDIE-grounded framework offers a **replicable, tool-agnostic blueprint** for large-scale AI integration
- 496 students · 505 original compositions — within six weeks
- Autonomous learning continued beyond the curriculum — **motivation genuinely internalized**
- Classroom output is **publication-ready** — verified end-to-end

19 Found My Sound



Before Q&A

The Story Behind the Story

 **The Joke:** "I'm 40 — same as your parents!"

Not Taken — three layers deep.

 **The Truth:** This study didn't start with my students.

It started with me —

A girl who always wanted to compose, but never could.

 **The Reveal:** 497 participants —
496 students + 1 teacher, still being scaffolded.

(that's me 😊)

Please bear with me.

Q & A

Questions & Discussion

Explore while you wait 🙌



Spotify
Found My Sound



Scaffold
Suno Prompt Site



Lyrics GPT
For SUNO



Student Work
Padlet



Materials
Course

“If questions go too deep, I may need AI to build me a scaffold right now.”